

[Sophia Exclusive] Grade 5A Workbook Error Remediation & Ch6 Review Mastery Worksheet

Student Name:
Sophia

Suggested Time: 45 ~ Total Points:
50 Mins 100

Target:

5A Targeted Remediation + Ch6 Review Blind Spots

💡 Pre-Test Pitfall Avoidance Rules:

① Circle keywords & units before writing; pair every quantity with its correct unit! ② Estimate before calculating; never invert operations! ③ Keep algebraic variables and constant numbers separate; always divide by 2 for triangle area!

Part 1: Conceptual Precision & Number Sense (Fill-in-the-Blanks & Comparison)

5 Questions, 20 Points

1. [Target ERR-SOP-13] Quotient Invariance & Decimal Point Scaling (4 pts)

Rule: When divisor stays constant, scaling the dividend scales the quotient equally. Small dividend / big divisor = quotient < 1 !

Given $48 \div 16 = 3$, write down the quotients directly using the rules of division:

(1) $4.8 \div 16 =$ _____ (2) $0.48 \div 16 =$ _____ (3) $0.048 \div 16 =$ _____

(4) $480 \div 16 =$ _____

2. [Target ERR-SOP-15] Comparing Operations with Decimals Less Than 1 (4 pts)

Rule: Multiplying by a number < 1 shrinks the value; dividing by a number < 1 enlarges the value!

Fill in each $\square$ with " $>$ ", " $<$ ", or " $=$ " without detailed calculation:

(1) $6.4 \times 0.8 \square 6.4 \div 0.8$ (2) $12.5 \div 0.5 \square 12.5 \times 0.5$ (3) $0.95 \times 1.02 \square 0.95 \div 1.02$

(4) $8.4 \div 1 \square 8.4 \times 1$

3. [Target ERR-SOP-18] Recurring Decimals & Remainder Cycles (4 pts)

Rule: Expand recurring digits to compare; use division remainder to pinpoint the Nth recurring digit!

(1) Compare: $0.714714 \square 0.\dot{7}1\dot{4}$ (Hint: expand and compare the 7th decimal place).

(2) The fraction $\frac{1}{7} = 0.\dot{1}4285\dot{7}$ has a repeating period of _____ digits. The 100th digit after the decimal point is _____.

4. [Target ERR-SOP-24] Generality of Algebraic Variables (4 pts)

Rule: Variables represent any value; never substitute specific constants into general formulas!

A square has side length a cm:

(1) Its perimeter is _____ cm; (2) Its area is _____ cm^2 ; (3) When $a = 6$, its area is _____ cm^2

5. [Target ERR-SOP-27] Multiplicative Comparison Sentence Structures (4 pts)

Rule: "How many times B is A?" means $A \div B$; identify the reference base carefully!

(1) "How many times 15 is 45?" Equation: _____, Result: _____

(2) "A car traveled 240 km, which is 3 times the distance a bike traveled." The bike's distance is _____ km.

Part 2: Algebraic Simplification & Sign Rules (True/False & Evaluation)

3 Questions, 15 Points

6. [Target ERR-SOP-21] Combining Like Terms vs. Constants (5 pts)

Rule: Only like terms with identical variables can be combined; never add a constant into variable coefficients!

True or False: The algebraic expression $4x + 7 + 2x$ simplifies to $13x$. (_____)

Write the correct simplification: $4x + 7 + 2x =$

Explain why:

7. [Target ERR-SOP-22] Additive Inverses & Sign Properties (5 pts)

Rule: Signs stay with their terms; $+y$ and $-y$ cancel out to 0, never to $-2y$!

True or False: The identity $x + x - y + y = 2x - 2y$ is always correct. (_____)

Write the standard step-by-step simplification: $x + x - y + y =$

8. [Target ERR-SOP-24] "Simplify First, Then Substitute" Discipline (5 pts)

Rule: Always simplify expressions completely before evaluating numerical values!

Given $y = 3.5$, evaluate the algebraic expression $15y - 7y + 20$ (**Direct substitution without simplifying is strictly prohibited**).

Step 1 (Combine like terms): $15y - 7y + 20 =$

Step 2 (Substitute value): When $y = 3.5$, Value =

Part 3: Rigorous Column Arithmetic & Mixed Operations**3 Questions, 20 Points****9. [Target ERR-SOP-14] Column Division with Decimal Divisors & Quotient Alignment (6 pts)**

Rule: Shift divisor decimal point to make it an integer; shift dividend identically; align quotient's decimal point!

Calculate: $197.6 \div 5.2$ (1) **Pre-calculation estimate:** Round 197.6 to _____, 5.2 to _____, estimated quotient $\approx$ _____.(2) **Column division** (clearly show shifted decimal points):

$$5.2 \overline{) 197.6}$$

Verification Check:

Is quotient an integer or decimal?

Final result =

10. [Target ERR-SOP-20] Triangle Area Formula & Rigor (6 pts)

Rule: Triangles and trapezoids must always be divided by 2; omitting division by 2 doubles the area!

A right triangle has legs of base $b = 0.36$ m and height $h = 0.25$ m.(1) Write the triangle area formula: $\text{Area} = \underline{\hspace{2cm}}$

(2) Calculate the area of this triangle (show step-by-step working and units):

Working:

Answer:

11. [Target ERR-SOP-26] Order of Operations with Decimals (8 pts)

Rule: Parentheses first, then multiplication/division from left to right; count decimal places carefully!

(1) $0.4 \times (3.2 - 0.8) \div 1.2$

(2) $7.4 \times 1.3 - 4.68 \div 0.9$

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Part 4: Word Problems & Keyword Modeling**4 Questions, 25 Points****12. [Target ERR-SOP-16] Two-Step Equal Sharing & Deducting Remainder (6 pts)**

Rule: When there is a remainder, subtract the remaining amount first before dividing into equal groups!

A bakery made a batch of cookies weighing 9.2 kg in total. The baker distributed them equally into 4 gift boxes, leaving 0.4 kg of cookies on the counter. How many kilograms of cookies were placed in each gift box?

Guided thinking: Total cookies actually packed into the boxes = _____ kg.

Equation & Working:

Answer:

13. [Target ERR-SOP-17] Multi-Variable Matching & Unit Alignment (6 pts)

Rule: Match unit price with corresponding quantity; never multiply kilograms by kilograms!

At a supermarket, apples cost 6.8 dollars per kg and bananas cost 4.5 dollars per kg. Ms. Wang bought 15 kg of apples and 24 kg of bananas for the school cafeteria. How much did she pay in total?

Equation & Working:

Answer:

14. [Target ERR-SOP-23] Average Score Total Delta Model (6 pts)

Rule: Total = Average $\times$ Number of Games; the difference in totals gives the score of the extra games!

The school girls' basketball team averaged 42 points in their first 6 tournament games. They then played 2 more games, raising their overall average across all 8 games to 45 points.

(1) How many total points did the team score in the last 2 games combined? (2) What was their average score in the last 2 games?

Step-by-step Working:

Answer: (1) Total points in last 2 games: _____ pts; (2) Average in last 2 games: _____ pts.

15. [Target ERR-SOP-19 & ERR-SOP-27] Sharing Pitfall & Fraction of the Whole (7 pts)

Rule: "Herself and 3 friends" = $1 + 3 = 4$ people! Fraction of whole uses 1 as whole, not number of slices!

"Emma bought a large pizza for lunch and cut it into 8 equal slices. She wants to share the pizza evenly between **herself** and **three** of her best friends."

(1) How many people are sharing the pizza altogether? _____ people.

(2) What **fraction** of the whole pizza will each person receive? _____ (fraction) = _____ (decimal).

(3) How many **slices** of pizza will each person get? _____ slices.

16. [Target 5A P44 ERR-SOP-25] Bar Model Total Length Conservation Equation (5 pts)

Rule: Equal ribbons mean equal total length; solve for x first, then equate the total length to find y !

The bar models below show three ribbons of identical total length: Ribbon 1 is $3x$, Ribbon 2 is $x + 8$, and Ribbon 3 is $4y$.

[Tier 1] |---- x ----|---- x ----|---- x ----| (Total $3x$)
 [Tier 2] |---- x ----|----- 8 -----| (Total $x + 8$)
 [Tier 3] |-- y --|-- y --|-- y --|-- y --| (Total $4y$)

(1) Since Ribbon 1 and Ribbon 2 have equal length, set up an equation in terms of x and solve:

Equation: _____ $\Rightarrow x =$ _____

(2) Find the total length of each ribbon, set up an equation for y , and solve:

Total length: _____ dm; Equation for y : _____ $\Rightarrow y =$ _____

17. [Target 5A P47 Work Collaboration] Road Paving Collaboration Problem (5 pts)

Engineering teams A and B work together to pave a highway measuring 125 km in total. Team A paves 3.5 km per day, and Team B paves 4.5 km per day. Team A worked alone for the first 6 days due to scheduling, after which Team B joined to work together. How many days will it take to finish the remaining highway after Team B joins?

Working:

Answer:

18. [Target 5A P49 Encounter Travel] Traveling Towards Each Other Equation (5 pts)

City A and City B are 540 km apart. A coach and a truck depart simultaneously from City A and City B respectively, traveling towards each other. They meet after 4.5 hours. The average speed of the coach is 64 km/h. Let the truck's speed be x km/h. **Set up an algebraic equation and solve for x .**

Let the truck's average speed be x km/h.

Equation:

Answer:

19. [Target 5A P51 Boat Rental Surplus & Deficit] Surplus and Shortage Algebraic Model (5 pts)

SUIS organized Grade 5 teachers and students for a boating activity in a park. If each boat holds 4 people, 6 people are left without a boat. If each boat takes 2 more people (i.e. 6 people per boat), exactly 1 boat is completely empty. How many boats were originally rented? How many total teachers and students participated?

Let the number of rented boats be x .

Total people in Scheme 1: _____; Total people in Scheme 2: _____

Equation & Solution:

Answer:

[Answer Key & Marking Rubric] (For Teacher Reference)

Part 1 (20 pts): 1. (1) 0.3 (2) 0.03 (3) 0.003 (4) 30 (1 pt each); 2. (1) < (2) > (3) < (4) = (1 pt each); 3. (1) < (2) pts (2) 6, 8 (1 pt each); 4. (1) $4a$ (2) a^2 (3) 36 (1+1+2 pts); 5. (1) $45 \div 15$, 3 (2 pts) (2) 80 (2 pts).

Part 2 (15 pts): 6. False (2 pts); Simplification: $6x + 7$ (2 pts); Reason: Only like terms can combine coefficients, constant 7 cannot merge with x (1 pt). 7. False (2 pts); Steps: $(x + x) + (-y + y) = 2x + 0 = 2x$ (3 pts). 8. Simplify: $8y + 20$ (2 pts); Evaluate: $8 \times 3.5 + 20 = 28 + 20 = 48$ (3 pts).

Part 3 (20 pts): 9. (1) Estimate: $200 \div 5 = 40$ (2 pts); (2) Column division: quotient is integer 38 (4 pts; deduct 3 pts if misplaced decimal point). 10. (1) $S = \frac{1}{2}bh$ (2 pts); (2) $0.36 \times 0.25 \div 2 = 0.045 \text{ m}^2$ (4 pts; deduct 3 pts if forgot to divide by 2). 11. (1) $0.4 \times 2.4 \div 1.2 = 0.96 \div 1.2 = 0.8$ (4 pts); (2) $9.62 - 5.2 = 4.42$ (4 pts).

Part 4 (25 pts): 12. Guide: 8.8 kg (1 pt); $(9.2 - 0.4) \div 4 = 2.2 \text{ kg}$ (4 pts); Statement (1 pt). 13. $6.8 \times 15 + 4.5 \times 24 = 102 + 108 = 210$ dollars (6 pts). 14. (1) $45 \times 8 - 42 \times 6 = 360 - 252 = 108$ pts (4 pts); (2) $108 \div 2 = 54$ pts (2 pts). 15. (1) 4 people (2 pts); (2) $1/4$, 0.25 (2 pts); (3) 2 slices (3 pts).

Part 5 (20 pts): 16. (1) $3x = x + 8 \implies 2x = 8 \implies x = 4$ (2 pts); (2) Total length 12 dm (1 pt), $4y = 12 \implies y = 3$ (2 pts). 17. Remaining: $125 - 3.5 \times 6 = 104 \text{ km}$; $104 \div (3.5 + 4.5) = 13$ days (5 pts). 18. Equation: $(64 + x) \times 4.5 = 540 \implies 64 + x = 120 \implies x = 56 \text{ km/h}$ (5 pts). 19. Expressions: $4x + 6$ and $6(x - 1)$ (2 pts); Equation: $4x + 6 = 6(x - 1) \implies 2x = 12 \implies x = 6$ boats (2 pts); Total people: $4 \times 6 + 6 = 30$ people (1 pt).